

Student's Name: _____

Problem 1: (2 + 2 = 4 points)

Consider a semiconductor that is uniformly doped with $N_d = 10^{14} \text{ cm}^{-3}$ and $N_a = 0$, with an applied electric field of $E = 100 \text{ V/cm}$. Assume that $\mu_n = 1000 \text{ cm}^2/\text{V-s}$ and $\mu_p = 0$. Also assume the following parameters:

$$\begin{aligned}N_c &= 2 \times 10^{19} (T/300)^{3/2} \text{ cm}^{-3} \\N_v &= 1 \times 10^{19} (T/300)^{3/2} \text{ cm}^{-3} \\E_g &= 1.10 \text{ eV}\end{aligned}$$

(a) Calculate the electric-current density at $T = 300 \text{ K}$.

Solution:

$$\begin{aligned}\text{(a)} \quad n_i^2 &= N_c N_v \exp\left(\frac{-E_g}{kT}\right) \\&= (2 \times 10^{19})(1 \times 10^{19}) \exp\left(\frac{-1.10}{0.0259}\right) = 7.18 \times 10^{19}\end{aligned}$$

or

$$n_i = 8.47 \times 10^9 \text{ cm}^{-3}$$

$$\text{For } N_d = 10^{14} \text{ cm}^{-3} \gg n_i \Rightarrow n_o = 10^{14} \text{ cm}^{-3}$$

Then

$$\begin{aligned}J &= \sigma E = e \mu_n n_o E \\&= (1.6 \times 10^{-19})(1000)(10^{14})(100)\end{aligned}$$

or

$$J = 1.60 \text{ A/cm}^2$$

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Problem 2: (2+2 = 4 pints)

The electron diffusion current density in a semiconductor is a constant and is given by $J_n = -2 \text{ A/cm}^2$. The electron concentration at $x = 0$ is $n(0) = 10^{15} \text{ cm}^{-3}$. (a) Calculate the electron concentration at $x = 20 \mu\text{m}$ if the material is silicon with $D_n = 30 \text{ cm}^2/\text{s}$. (b) Repeat part (a) if the material is GaAs with $D_n = 230 \text{ cm}^2/\text{s}$.

{Hint: note that $\frac{dn}{dx} = \frac{\Delta n}{\Delta x} = \frac{n(x_0) - n(x_1)}{x_0 - x_1}$ you are asked to find value of $n(x_1)$

where x_1 is $20 \mu\text{m}$ }

Solution:

$$\begin{aligned} \text{(a)} \quad J_n &= eD_n \frac{dn}{dx} = eD_n \frac{\Delta n}{\Delta x} \\ -2 &= (1.6 \times 10^{-19})(30) \left[\frac{10^{15} - n(x_1)}{0 - 20 \times 10^{-4}} \right] \\ 4 \times 10^{-3} &= 4.8 \times 10^{-3} - 4.8 \times 10^{-18} n(x_1) \end{aligned}$$

which yields

$$n(x_1) = 1.67 \times 10^{14} \text{ cm}^{-3}$$

$$\begin{aligned} \text{(b)} \quad -2 &= (1.6 \times 10^{-19})(230) \left[\frac{10^{15} - n(x_1)}{0 - 20 \times 10^{-4}} \right] \\ 4 \times 10^{-3} &= 3.68 \times 10^{-2} - 3.68 \times 10^{-17} n(x_1) \end{aligned}$$

$$n(x_1) = 8.91 \times 10^{14} \text{ cm}^{-3}$$

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Problem 3: (2+2+1= 5 pints)

An n-type silicon sample contains a donor concentration of $N_d = 10^{16} \text{ cm}^{-3}$. The minority carrier hole lifetime is found to be $\tau_{p0} = 20 \mu\text{s}$. (a) What is the lifetime of the majority carrier electrons? (b) Determine the thermal-equilibrium generation rate for electrons and holes in this material. (c) Determine the thermal-equilibrium recombination rate for electrons and holes in this material.

{Hint: Recombination rates are equal $\gg \gg \frac{n_o}{\tau_{nO}} = \frac{p_o}{\tau_{pO}}$ }

Solution:

(a) Recombination rates are equal,

$$\frac{n_o}{\tau_{nO}} = \frac{p_o}{\tau_{pO}}$$

$$n_o = N_d = 10^{16} \text{ cm}^{-3}$$

$$p_o = \frac{n_i^2}{n_o} = \frac{(1.5 \times 10^{10})^2}{10^{16}} = 2.25 \times 10^4 \text{ cm}^{-3}$$

Then

$$\frac{10^{16}}{\tau_{nO}} = \frac{2.25 \times 10^4}{20 \times 10^{-6}}$$

which yields

$$\tau_{nO} = 8.89 \times 10^{-6} \text{ s}$$

(b) Generation rate = recombination rate

$$G = \frac{2.25 \times 10^4}{20 \times 10^{-6}} = 1.125 \times 10^9 \text{ cm}^{-3} \text{ s}^{-1}$$

(c)

$$R = G = 1.125 \times 10^9 \text{ cm}^{-3} \text{ s}^{-1}$$

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Problem 4: (2.5 + 2.5 + 2 = 7 pints)

A Si sample with $10^{16}/\text{cm}^3$ (n_0) is excited by a laser such that it generates (G) $10^{19} / \text{cm}^3$ electron hole pairs (EHPs) per second uniformly in the sample. The laser to cause the sample to heat up to 450 K. Find the position of Fermi level (called quasi-Fermi level) and change in the conductivity of the sample upon shining the light. The electron and hole both life-times (τ) are 10^{-5} sec. and $D_p = 12 \text{ cm}^2/\text{s}$; $D_n = 36 \text{ cm}^2/\text{s}$ and $n_i = 10^{14} \text{ cm}^{-3}$ at 450K. Draw the energy band diagram after shining the light. {Hint: $n = n_0 + \delta n$, $p = p_0 + \delta p$ and $\delta n = \delta p = G \cdot \tau$, once you have found n and p values you can calculate $F_n - E_{Fi}$ and $E_{Fi} - F_p$, is asked for first part of the question, to calculate change in conductivity you first find mobility for electron and holes using $kT\mu_n/q = D_n$ and $kT\mu_p/q = D_p$ and the change in conductivity will be $\Delta\sigma = q [\mu_n \cdot \delta n + \mu_p \cdot \delta p]$.

Find the separation of the quasi-Fermi levels and the change of conductivity when shining light.

The light induced electron-hole pair concentration is determined by:

$$\delta n = \delta p = g_{op} \cdot \tau = 10^{19} \frac{1}{\text{cm}^3 \cdot \text{s}} \cdot 10^{-5} \text{s} = 10^{14} \frac{1}{\text{cm}^3}$$

$$\delta n < \text{dopant concentration of } n_0 = 10^{16} \frac{1}{\text{cm}^3} \text{ so low level}$$

$$n = n_0 + \delta n = 10^{16} \frac{1}{\text{cm}^3} + 10^{14} \frac{1}{\text{cm}^3} \approx 10^{16} \frac{1}{\text{cm}^3}$$

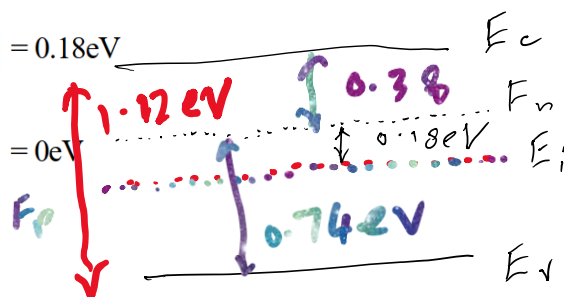
$$p = p_0 + \delta p = \frac{n_i^2}{n_0} + \delta p = \frac{(10^{14} \frac{1}{\text{cm}^3})^2}{10^{16} \frac{1}{\text{cm}^3}} + 10^{14} \frac{1}{\text{cm}^3} \approx 10^{14} \frac{1}{\text{cm}^3}$$

$$kT \text{ for } 450\text{K} = 0.0259\text{eV} \cdot \frac{450\text{K}}{300\text{K}} = 0.039\text{eV}$$

$$F_n - E_i = kT \cdot \ln\left(\frac{n}{n_i}\right) = 0.039\text{eV} \cdot \ln\left(\frac{10^{16} \frac{1}{\text{cm}^3}}{10^{14} \frac{1}{\text{cm}^3}}\right) = 0.18\text{eV}$$

$$E_i - F_p = kT \cdot \ln\left(\frac{p}{n_i}\right) = 0.039\text{eV} \cdot \ln\left(\frac{10^{14} \frac{1}{\text{cm}^3}}{10^{14} \frac{1}{\text{cm}^3}}\right) = 0\text{eV}$$

$$F_n - F_p = 0.18\text{eV}$$



$$\mu_n = \frac{D_n}{\frac{kT}{q}} = \frac{36 \frac{\text{cm}^2}{\text{s}}}{0.039\text{V}} = 927 \frac{\text{cm}^2}{\text{V} \cdot \text{s}}$$

$$\mu_p = \frac{D_p}{\frac{kT}{q}} = \frac{12 \frac{\text{cm}^2}{\text{s}}}{0.039\text{V}} = 309 \frac{\text{cm}^2}{\text{V} \cdot \text{s}}$$

$$\Delta\sigma = q \cdot (\mu_n \cdot \delta n + \mu_p \cdot \delta p) = 1.6 \cdot 10^{-19} \text{C} \cdot (927 \frac{\text{cm}^2}{\text{V} \cdot \text{s}} \cdot 10^{14} \frac{1}{\text{cm}^3} + 309 \frac{\text{cm}^2}{\text{V} \cdot \text{s}} \cdot 10^{14} \frac{1}{\text{cm}^3}) = 0.0198 \frac{1}{\Omega \cdot \text{cm}}$$